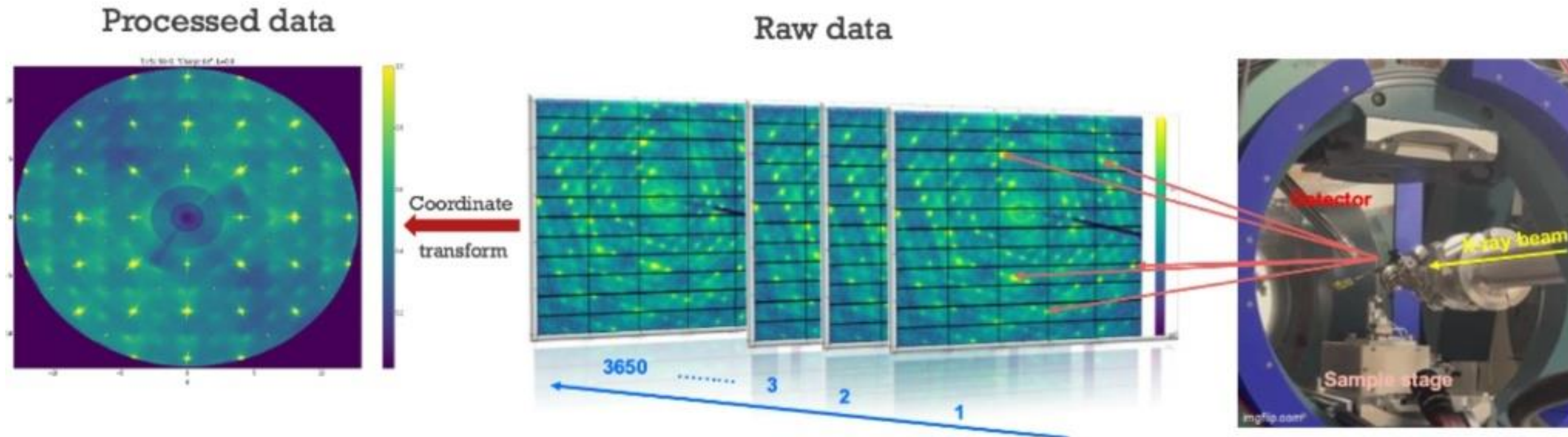


Background

X-Ray Scattering at CHESS

- The Q-Mapping for Quantum Materials (QM2) beamline generates large, high-dimensional reciprocal-space datasets across temperature sweeps to study quantum materials.
- Raw data is processed into 3D arrays of scattering intensity across reciprocal space and temperature.



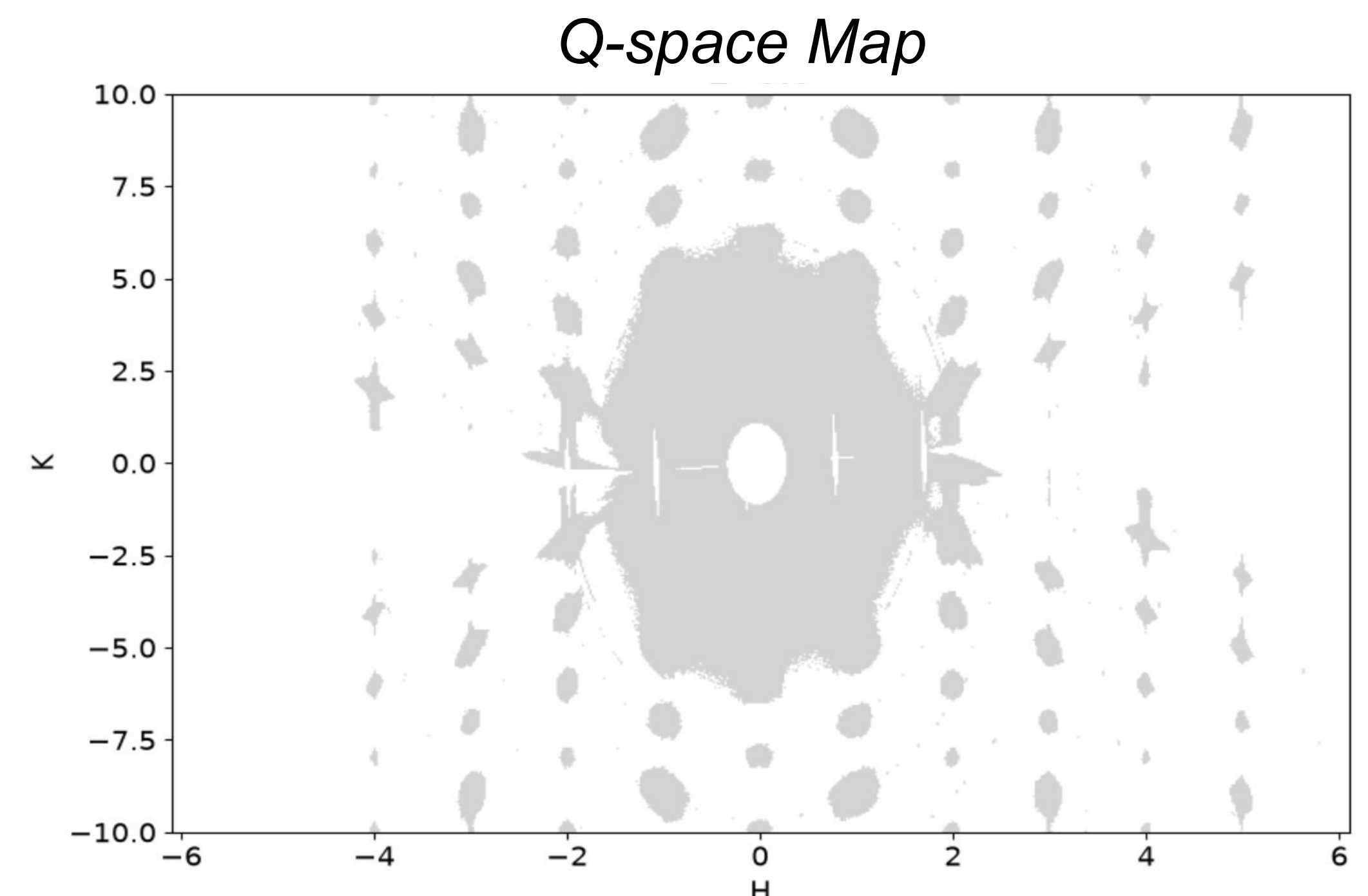
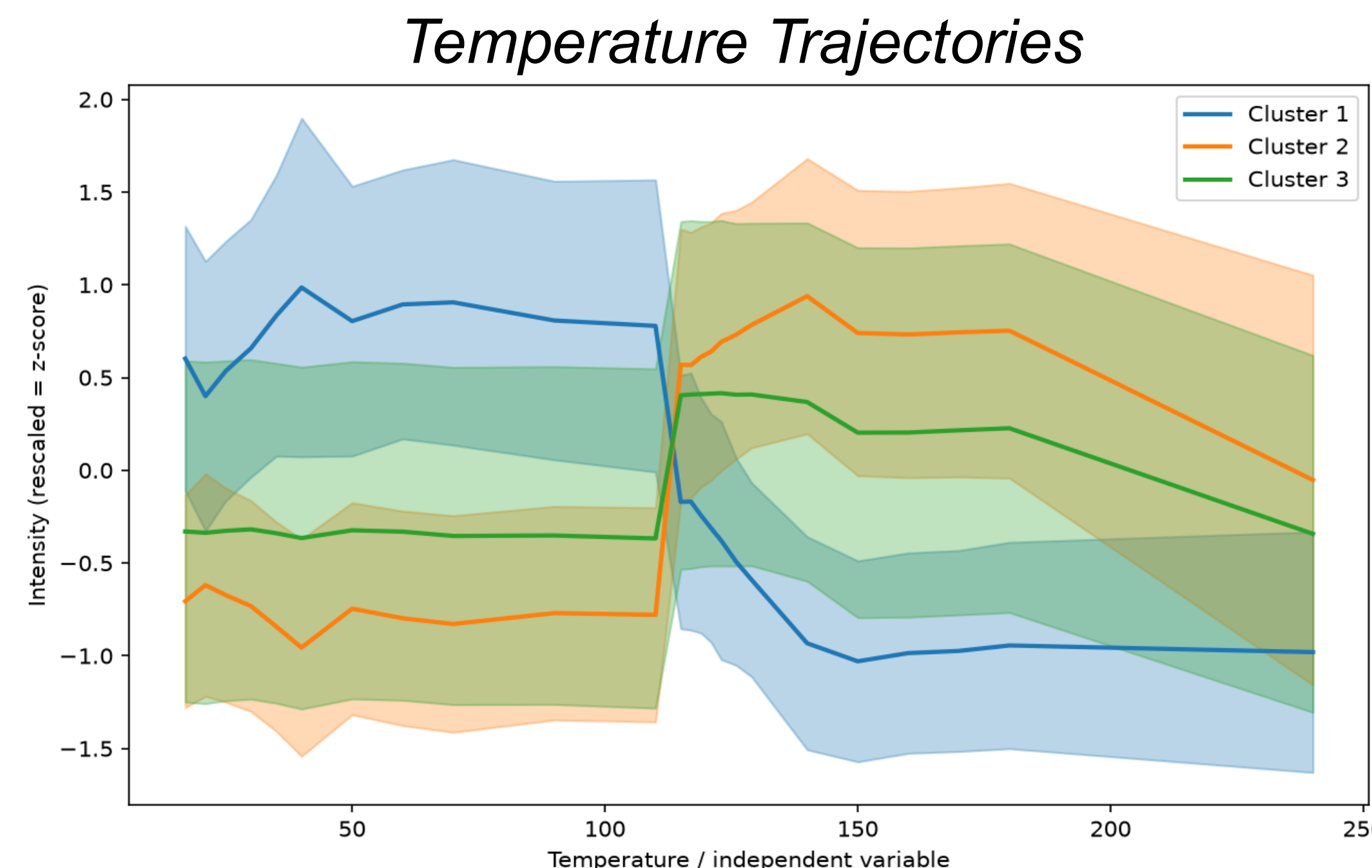
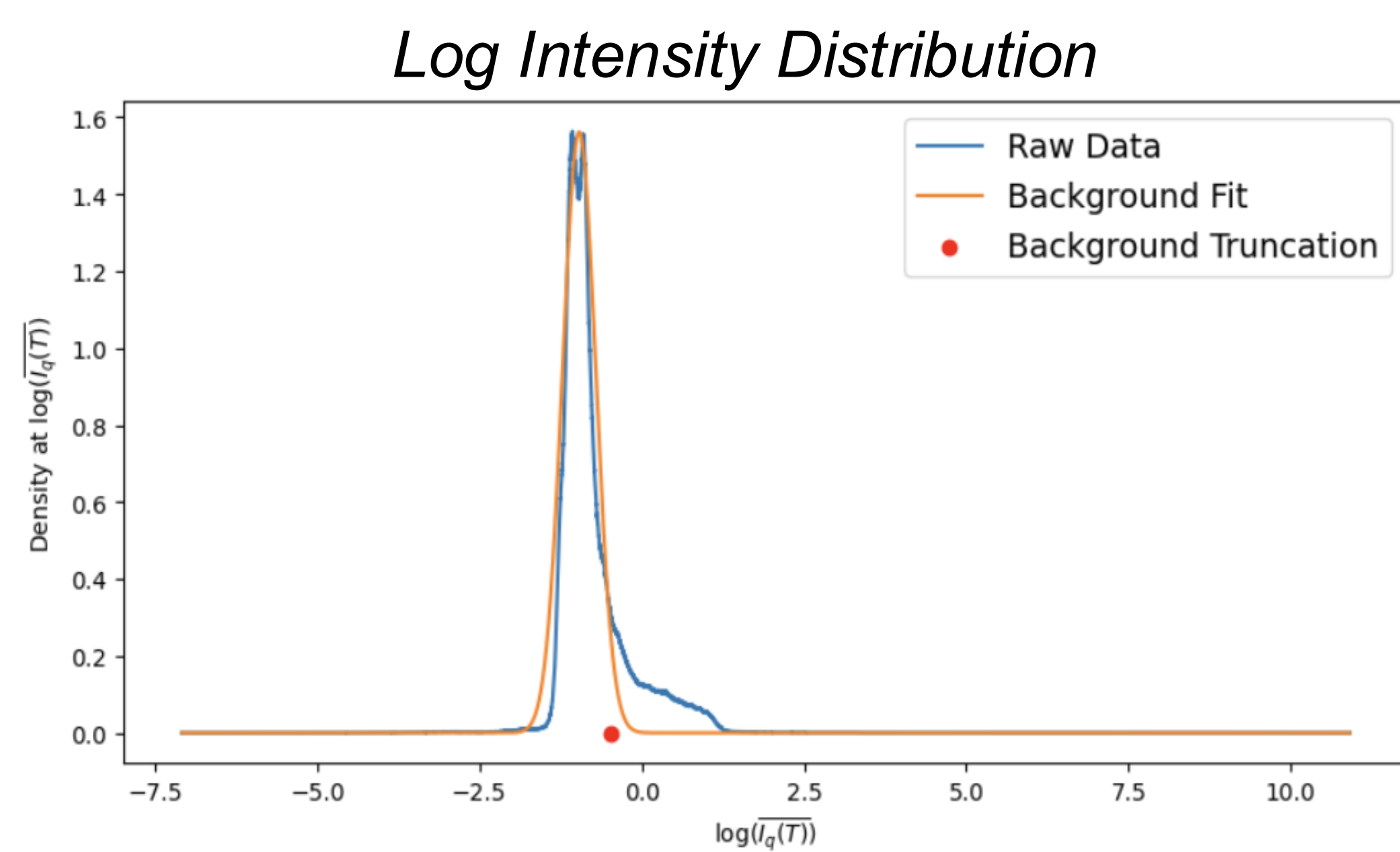
QM2 Experimental & Data Pipeline (Adapted from S. Sarker)

Unsupervised Machine Learning with X-TEC¹

- Uses Gaussian Mixture Models to cluster pixels by temperature trajectories.
- Identifies phase transitions and order parameters without prior crystal models.

Summer Goals

- Goal 1: Build a reproducible pipeline for GPU hyperparameter sweeps and judge evaluations.
- Goal 2: Test if automatic cluster scoring can match expert judgement.

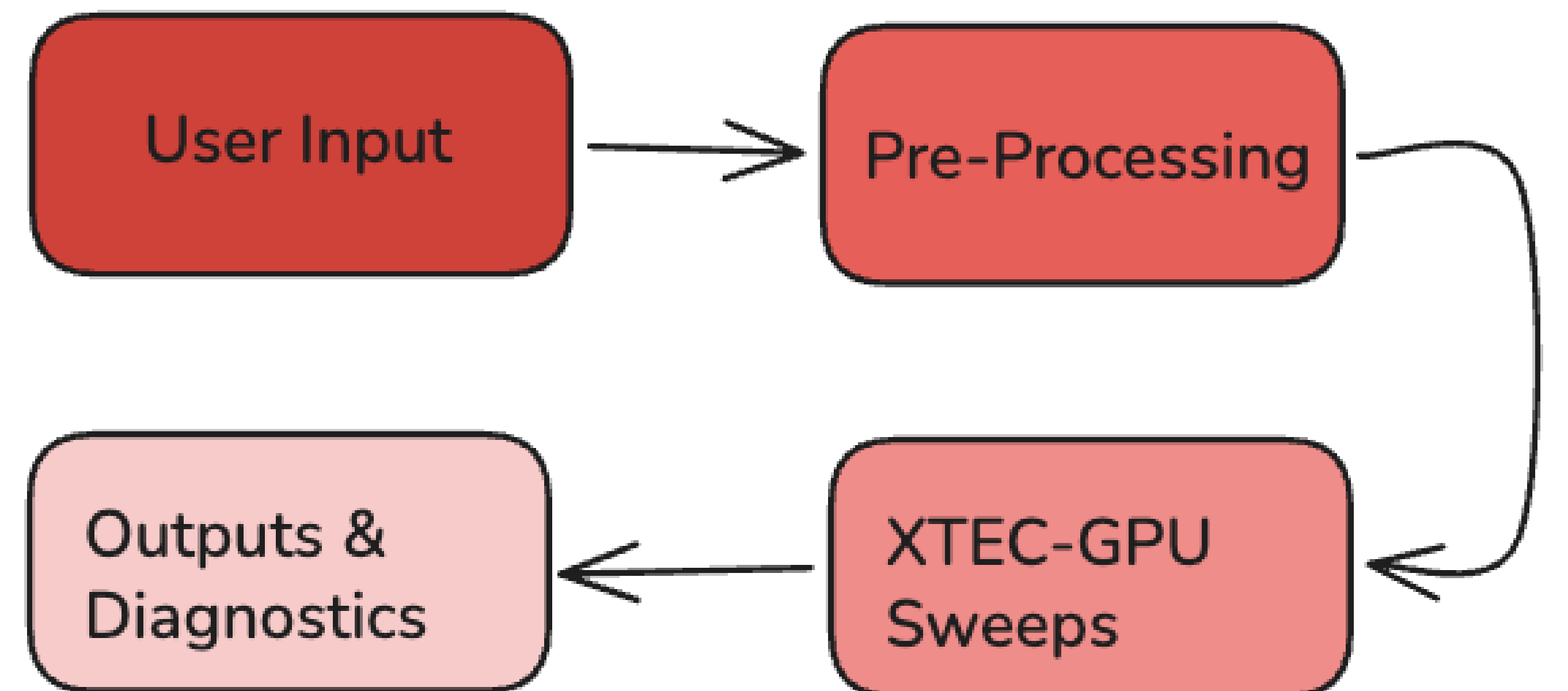


Goal 1: Automating The Pipeline

Unified Workflow Control

- Configuration cell controls paths, parameters, scaling, and execution.
- Auto-detects temperatures, handles varied grid dimensions, and filters noise.
- Runs XTEC-GPU hyperparameter sweeps directly.

Automated Pipeline Workflow



Diagnostics

- Exports and saves temperature trajectories, Q-space maps, and recommendations.

Validation & Research Impact

- Tested on datasets like Ta₂NiSeS₇, generalizing across grid sizes and symmetries without code edits.
- Standardizes analysis across CHESS beamline runs.

Goal 2: Matching Expert Judgement

The Problem

- Hand-chosen metric weights fail to generalize across materials.
- Rescaling methods (mean and Z-score) change recommendations, preventing meaningful score comparisons.

Approach

- Performed systematic weight sweeps to measure impact on recommendations.
- Evaluated new metrics (BIC elbow, amplitude, signal-to-noise, redundancy, and Q-space compactness).
- Trained a Bradley-Terry model on synthetic datasets to learn optimal weights.

Results & Key Findings

- Weight sweeps revealed only 1 of 6 original weights impacted recommendations (2 had zero effect).
- No single metric or learned combination consistently agreed with expert judgement.
- Key Finding:** Current metrics miss crystal symmetry, the primary factor experts use to evaluate cluster quality.

Future Work

- Develop metrics that encode crystal symmetry and re-evaluate the judge via our Bradley-Terry framework.
- Expand testing to confirm pipeline generalizability across materials.

References

- J. Venderley et al., Proc. Natl. Acad. Sci. U.S.A. 119, 24, e2109665119 (2022).

Acknowledgements

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Scan to view XTEC-Pipeline Github